The documentation and process conversion measures necessary to comply with this document shall be completed by 22 May 2022.

INCH-POUND

MIL-PRF-19500/313K
w/AMENDMENT 1
22 February 2022
SUPERSEDING
MIL-PRF-19500/313K
19 August 2016

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, NPN, SILICON, LOW POWER, TYPES 2N2432,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for low power, high speed chopper, NPN silicon transistors. Four levels of product assurance (JAN, JANTX, JANTXV and JANS) are provided for each device type as specified in MIL-PRF-19500. Two levels of product assurance (JANHC and JANKC) are provided for die.

1.2 Package outlines. The device packages for the encapsulated device types are as follows: (TO-18) in accordance with figure 1, (JANHC/JANKC die) in accordance with figure 2, and (UB package) in accordance with figure 3.

1.3 Maximum ratings, unless otherwise specified, $T_C$ =+ 25°C.

| Type | $V_{CBO}$ V dc | $V_{CEO}$ V dc | $V_{ECO}$ V dc | $I_C$ mA dc | $T_J$ and $T_{STG}$ °C |
|---|---|---|---|---|---|
| 2N2432, 2N2432UB | 30 | 30 | 15 | 100 | -65 to +200 |
| 2N2432A, 2N2432AUB | 45 | 45 | 18 | 100 | |

| Type | $P_T$ $T_A$ = +83°C (1) | $P_T$ $T_C$ = +150°C (1) | $P_T$ $T_{SP}$ = +165°C | $R_{\theta JA}$ (2) | $R_{\theta JC}$ (2) | $R_{\theta JSP}$ (2) |
|---|---|---|---|---|---|---|
| | mW | mW | mW | °C/W | °C/W | °C/W |
| 2N2432 | 360 | 360 | N/A | 325 | 150 | N/A |
| 2N2432A | 360 | 360 | N/A | 325 | 150 | N/A |
| 2N2432UB | N/A | N/A | 360 | N/A | N/A | 95 |
| 2N2432AUB | N/A | N/A | 360 | N/A | N/A | 95 |

(1) For derating, see figure 4 and figure 5.
(2) For thermal impedance curves, see figure 6, figure 7, and figure 8.

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1.4 Primary electrical characteristics at $T_A$ = +25°C.

| Limits | $h_{FE1}$ $V_{CE}$ = 5 V dc | $h_{FE2}$ $V_{CE}$ = 5 V dc | $h_{FE(inv)1}$ $V_{CE}$ = 5 V dc $I_E$ = 200 µA dc | | $V_{CE(sat)}$ $I_C$ = 10 mA dc | $r_{ec(on)}$ $I_e$ = 100 µA ac (rms) $I_B$ = 1 mA dc, $I_E$ = 0, f = 1 kHz | |
|---|---|---|---|---|---|---|---|
| | $I_C$ = 10 µA dc | $I_C$ = 1 mA dc | 2N2432 | 2N2432A | $I_B$ = 500 µA dc | 2N2432 2N2432UB | 2N2432A 2N2432AUB |
| Min | 30 | 80 | 2 | 3 | V dc | Ohms | Ohms |
| Max | | 400 | | | .15 | 20 | 15 |

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.5 for PIN construction example, 6.6 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX" “JANTXV”, and "JANS".

1.5.1.2 Quality level designators for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.2.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "2432".

1.5.3 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

| | |
|---|---|
| | A blank first suffix symbol indicates a standard voltage, axial encapsulated device (see figure 1 ). |
| A | An “A” first suffix symbol Indicates higher $V_{CBO}$, $V_{CEO}$, and $V_{ECO}$ voltage. |
| UB | Indicates a surface mount (2N2432UB) (see figure 3) |

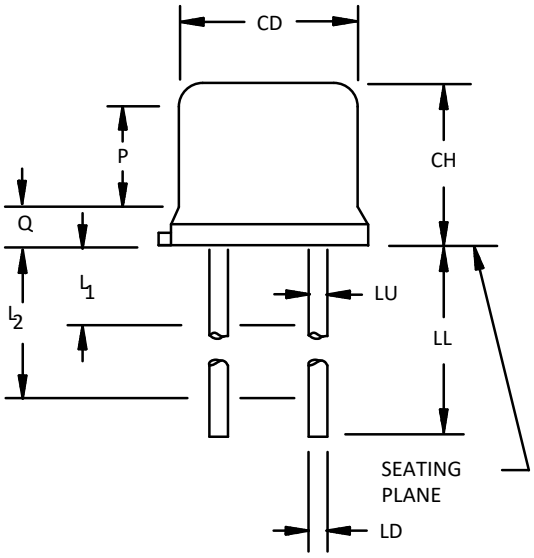
1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QML-19500.

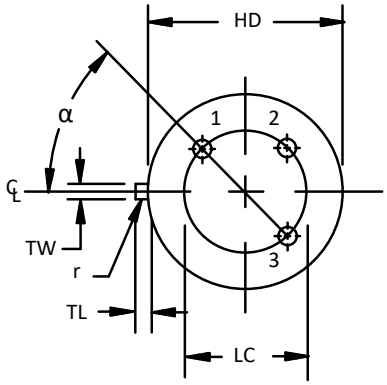
1.5.5 Die identifiers for unencapsulated device (manufacturers and critical interface identifiers). The manufacturer die identifier that is applicable for this specification sheet is "B" (see figure 2, 6.5, and 6.6).



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| Symbol | in | in | mm | mm | Note |
|---|---|---|---|---|---|
| | Min | Max | Min | Max | |
| CD | .178 | .195 | 4.52 | 4.95 | |
| CH | .170 | .210 | 4.32 | 5.33 | |
| HD | .209 | .230 | 5.31 | 5.84 | |
| LC | .100 TP | | 2.54 TP | | 6 |
| LD | .016 | .021 | 0.41 | 0.53 | 7,8 |
| LL | .500 | .750 | 12.70 | 19.05 | 7,8 |
| LU | .016 | .019 | 0.41 | 0.48 | 7,8 |
| L1 | | .050 | | 1.27 | 7,8 |
| L2 | .250 | | 6.35 | | 7,8 |
| P | .100 | | 2.54 | | |
| Q | | .030 | | 0.76 | 5 |
| TL | .028 | .048 | 0.71 | 1.22 | 3,4 |
| TW | .036 | .046 | 0.91 | 1.17 | 3 |
| r | | .010 | | 0.25 | 10 |
| α | 45° TP | | 45° TP | | 6 |



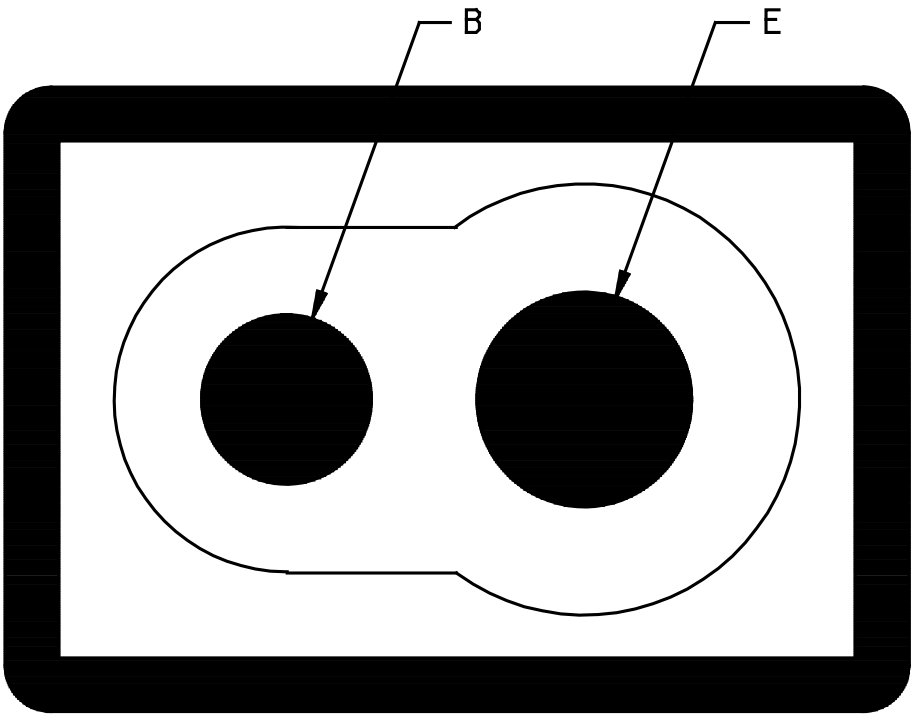




NOTES:
1. Dimension are in inches.
2. Millimeters are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC.
7. Dimension LU applies between $L_1$ and $L_2$. Dimension LD applies between $L_2$ and LL minimum. Diameter is uncontrolled in $L_1$ and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
* 11. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-18).



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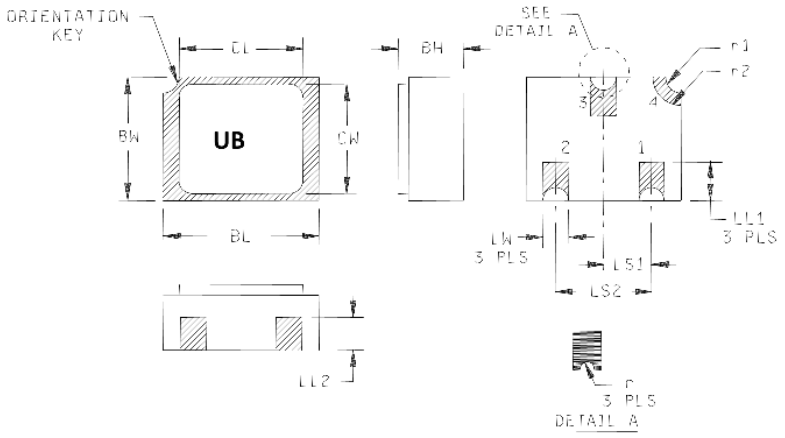

NOTES:

| | | |
|---|---|---|
| 1. | Chip size | .015 x .019 inch ±.001 inch, (0.38 x 0.48 ±0.02 millimeter). |
| 2. | Chip thickness | .010 ±.0015 inch, (0.25 ±0.04 millimeter). |
| 3. | Top metal | Aluminum 15,000Å minimum, 18,000Å nominal. |
| 4. | Back metal | Gold 3,500Å minimum, 5,000Å nominal. |
| 5. | Backside | Collector |
| 6. | Bonding pad | B = .003 inch (0.08 millimeter).<br>E = .004 inch diameter (0.10 millimeter). |
| 7. | Passivation | $Si_3N_4$ (Silicon Nitride) 5.6 kÅ min, 8 kÅ nom. |
| 8. | See 6.4. | |

FIGURE 2. Physical dimensions, JANHC and JANKC (B - version) die.



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| Symbol | in | in | mm | mm | Note | Symbol | in | in | mm | mm | Note |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Max | Min | Max | | | Min | Max | Min | Max | |
| BH | .046 | .056 | 1.17 | 1.42 | | LS1 | .035 | .039 | 0.89 | 0.99 | |
| BL | .115 | .128 | 2.92 | 3.25 | | LS2 | .071 | .079 | 1.80 | 2.01 | |
| BW | .085 | .108 | 2.16 | 2.74 | | LW | .016 | .024 | 0.41 | 0.61 | |
| CL | .115 | .128 | 2.92 | 3.25 | | r | | .008 | | 0.20 | |
| CW | .085 | .108 | 2.16 | 2.74 | | r1 | | .012 | | 0.31 | |
| LL1 | .022 | .038 | 0.56 | 0.96 | | r2 | | .022 | | 0.56 | |
| LL2 | .017 | .035 | 0.43 | 0.89 | | | | | | | |

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
* 5. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions, surface mount (UB version).



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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

| | |
|---|---|
| $h_{FE(inv)}$ | Forward current transfer ratio except that the collector and emitter shall be interchanged. |
| $I_e$ | Emitter current (rms). |
| PCB | Printed circuit board. |
| $r_{ec(on}$ | Small signal emitter to collector on state resistance. |
| $R_{\theta JA}$ | Thermal resistance junction to ambient. |
| $R_{\theta JC}$ | Thermal resistance junction to case. |
| $R_{\theta JSP}$ | Thermal resistance junction to solder pads. |
| $V_{BC}$ | Base to collector voltage. |
| $V_{(BR)ECO}$ | Breakdown voltage, emitter to collector, with base open circuited. |
| $V_{EC(ofs)}$ | Emitter to collector offset voltage, i.e., open circuit voltage between emitter and collector when the base to collector junction is forward biased. |
| $V_{ec}$ | Emitter to collector voltage (rms). |



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3.4 Interface requirements and physical dimensions. Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, figure 2, and figure 3.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Low power transistor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and table I and table II and 4.5.7).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC devices. JANHC and JANKC devices shall be qualified in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.



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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen | Measurement | Measurement |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| (1) 3c | Thermal impedance, method 3131 of MIL-STD-750, see 4.3.3. | Thermal impedance, method 3131 of MIL-STD-750, see 4.3.3. |
| 7 | Optional | Optional |
| 9 | $I_{CBO2}$, $h_{FE1}$ | Not applicable |
| 10 | 48 hours minimum | 48 hours minimum |
| 11 | $I_{CBO2}$; $h_{FE1}$;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater. $\Delta h_{FE1}$ = ±15 percent. | $I_{CBO2}$, $h_{FE1}$ |
| 12 | See 4.3.1 | See 4.3.1 |
| 13 | Subgroups 2 and 3 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE1}$ = ±15 percent. | Subgroup 2 of table I herein;<br>$\Delta I_{CBO2}$ = 100 percent of initial value or 5 nA dc, whichever is greater;<br>$\Delta h_{FE1}$ = ±15 percent. |
| 14 | Required | Required |

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CB}$ = 10 - 30 V dc; power shall be applied using a minimum $P_D$ = 100 percent of $P_T$ maximum, $T_A$ ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, $T_J$, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-PRF-19500 "Discrete Semiconductor Die/Chip Lot Acceptance". Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.



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* 4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining $I_M$, $I_H$, $t_H$, $t_{MD}$ (and $V_C$ where appropriate). The thermal impedance limit shall comply with the thermal impedance graph in figure 6, figure 7, and figure 8, (less than or equal to the curve value at the same $t_H$ time) or shall be less than the process determined statistical maximum limit as outlined in method 3131 of MIL-STD-750. See table II, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.7 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with table I, subgroup 2 and 4.5.7 herein.

4.4.2.1 Group B inspection (JANS), table E-VIa of MIL-PRF-19500.

| | Subgroup | Method | Condition |
|---|---|---|---|
| * | B4 | 1037 | $V_{CB}$ = 10 - 30 V dc. |
| * | B5 | 1027 | $V_{CB}$ = 10 - 30 V dc; $P_D \geq$ 100 percent of maximum rated $P_T$ (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.)<br><br>Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table VIa, adjust $T_A$ to achieve $T_J$ = +275°C minimum.<br><br>Option 2: 216 hours minimum, sample size = 45, c = 0; adjust $T_A$ to achieve a $T_J$ = +225°C minimum. |

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL-PRF-19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failures mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

| Step | Method | Condition |
|---|---|---|
| 1 | 1026 | Steady-state life: 1,000 hours minimum, $V_{CB}$ = 10 - 30 V dc, power shall be applied and adjust $T_A$ to achieve $T_J$ = +150°C minimum using a minimum of $P_D$ = 100 percent of maximum rated $P_T$ as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test Time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours. |
| 2 | 1048 | Blocking life, $T_A$ = +150°C, $V_{CB}$ = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0. |
| 3 | 1032 | High-temperature life (non-operating), t = 340 hours, $T_A$ = +200°C. n = 22, c = 0. |



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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

* b. Must be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, subgroup 2 and 4.5.7.

4.4.3.1 Group C inspection (JANS), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; (not applicable for UB devices). |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves. |
| C6 | 1026 | 1,000 hours at $V_{CB}$ = 10 - 30 V dc; power shall be applied to achieve $T_J$ = +150°C minimum and a minimum of $P_D$ = 100 percent of maximum rated $P_T$ as defined in 1.3, n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours, adjust $T_A$. |

* 4.4.3.2 Group C inspection (JAN, JANTX, and JANTXV), table E-VII of MIL-PRF-19500.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | Test condition E; not applicable for UB devices. |
| C5 | 3131 | $R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3). |
| C6 | | Not applicable. |

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; delta measurements shall be in accordance with the applicable steps of 4.5.7.



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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Emitter to collector cutoff current. Method of test shall be in accordance with method 3041 of MIL-STD-750, test condition C, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.3 Emitter to collector breakdown voltage. Method of test shall be in accordance with method 3011 of MIL-STD-750, test condition D, except that all references to the collector and emitter of the transistor under test shall be interchanged.

4.5.4 Forward current transfer ratio (inverted connection). Method of test shall be in accordance with method 3076 of MIL-STD-750, except that all references to the collector and emitter of the transistor under test shall be interchanged. Then $h_{FE(inv)} = I_E / I_B$.

4.5.5 Emitter to collector offset voltage. The transistor shall be tested in the circuit on figure 9. The base current shall be adjusted to the specified value. The voltage between the emitter and collector shall be measured using a voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement.

4.5.6 Small signal emitter to collector on-state resistance. The transistor shall be tested in the circuit of figure 10. The base current shall be adjusted to the specified value and an ac sinusoidal signal current, $I_e$, of the specified rms value shall be applied between the emitter and collector. The rms voltage, $V_{ec}$, between the emitter and collector shall be measured using an ac voltmeter with an input impedance high enough that halving it does not change the measured value within the required accuracy of the measurement. The small signal emitter to collector on-state resistance shall then be determined as follows:

$$r_{ec(on)} = V_{ec} / I_E$$

Where $V_{ec}$ is the rms voltage between the emitter and collector.

4.5.7 Delta requirements. Delta requirements shall be as specified below: (1) (2) (3) (4)

| Step | Inspection | MIL-STD-750 | | Symbol | Limit | Unit |
|---|---|---|---|---|---|---|
| | | Method | Conditions | | | |
| 1 | Collector-base cutoff current | 3036 | Bias condition D | $\Delta I_{CB02}$ | 100 percent of initial value or 5 nA dc, whichever is greater. | |
| | 2N2432, 2N2432UB | | $V_{CB}$ = 25 V dc | | | |
| | 2N2432A, 2N2432AUB | | $V_{CB}$ = 40 V dc | | | |
| 2 | Forward current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1.0 mA dc; pulsed see 4.5.1 | $\Delta h_{FE2}$ | ±25 percent change from initial reading. | |

(1) Devices which exceed the table I limits for this test shall not be accepted.
(2) The delta measurements for group B, table E-VIa (JANS) of MIL-PRF-19500 are as follows: Subgroup 3 and 5; see 4.5.7 herein, steps 1 and 2.
(3) The delta measurements for 4.4.2.2 herein (group B, JAN, JANTX, and JANTXV) are as follows: Steps 1 and 2 of 4.5.7 shall be performed after each step in 4.4.2.2 herein.
(4) The delta measurements for group C, table E-VII of MIL-PRF-19500 are as follows: Subgroup 6, see 4.5.7 herein, steps 1 and 2 for JANS only.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 2/ | | | | | | |
| Visual and mechanical examination 3/ | 2071 | | | | | |
| Solderability 3/ 4/ | 2026 | n = 15 leads, c = 0 | | | | |
| Resistance to solvents 3/ 4/ 5/ | 1022 | n = 15 devices, c = 0 | | | | |
| Salt atmosphere 4/ | 1041 | (Laser marked devices only) n = 6 devices, c = 0. | | | | |
| Temp cycling 3/ 4/ | 1051 | Test condition C, 25 cycles. n = 22 devices, c = 0 | | | | |
| Heremetic seal 4/ | 1071 | n = 22 devices, c = 0 | | | | |
| Fine leak<br>Gross leak<br><br>Electrical measurements 4/ | | Table I, subgroup 2 | | | | |
| Bond strength 3/ 4/ | 2037 | Precondition $T_A$ = +250°C at t = 24 hrs or $T_A$ = +300°C at<br><br>t = 2 hrs n = 11 wires, c = 0 | | | | |
| Decap internal visual design verification 4/ | 2075 | n = 4, c = 0. | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance 6/ | 3131 | See 4.3.3 | $Z_{\theta JX}$ | | | °C/W |
| Breakdown voltage collector to emitter | 3011 | Bias condition D; $I_C$ = 10 mA dc; pulsed (see 4.5.1). | $V_{(BR)CEO}$ | | | |
| 2N2432, 2N2432UB | | | | 30 | | V dc |
| 2N2432A, 2N2432AUB | | | | 45 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO1}$ | | | |
| 2N2432, 2N2432UB | | $V_{CB}$ = 30 V dc | | | 100 | µA dc |
| 2N2432A, 2N2432AUB | | $V_{CB}$ = 45 V dc | | | 100 | µA dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Breakdown voltage emitter to collector | 3011 | Bias condition D; $I_E$ = 100 µA dc; $I_B$ = 0 (see 4.5.3). | $V_{(BR)ECO1}$ | | | |
| 2N2432, 2N2432UB | | | | 15 | | V dc |
| 2N2432A, 2N2432AUB | | | | 18 | | V dc |
| Breakdown voltage emitter to collector | 3011 | Bias condition D; $I_E$ = 10 mA dc; $I_B$ = 0; pulsed (see 4.5.1 and 4.5.3) | $V_{(BR)EC02}$ | 10 | | V dc |
| Collector to base cutoff current | 3036 | Bias condition D | $I_{CBO2}$ | | | |
| 2N2432, 2N2432UB | | $V_{CE}$ = 25 V dc | | | 10 | nA dc |
| 2N2432A, 2N2432AUB | | $V_{CE}$ = 40 V dc | | | 10 | nA dc |
| Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES1}$ | | | |
| 2N2432, 2N2432UB | | $V_{CB}$ = 25 V dc | | | 10 | nA dc |
| 2N2432A, 2N2432AUB | | $V_{CB}$ = 40 V dc | | | 10 | nA dc |
| Emitter to collector cutoff current | 3041 | Bias condition C; $V_{EC}$ = 15 V dc; $V_{BC}$ = 0; (see 4.5.2) | $I_{ECS1}$ | | 2 | nA dc |
| Emitter to base cutoff current | 3061 | Bias condition D; $V_{EB}$ = 15 V dc | $I_{EBO}$ | | 2 | nA dc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 10 µA dc | $h_{FE1}$ | 30 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc | $h_{FE2}$ | 80 | 400 | |
| Forward-current transfer ratio (inverted connection) | 3076 | $V_{CE}$ = 5 V dc; $I_E$ = 0.2 mA dc; (see 1.4 and 4.5.4) | $h_{FE(inv)1}$ | | | |
| 2N2432, 2N2432UB | | | | 2 | | |
| 2N2432A, 2N2432AUB | | | | 3 | | |
| Saturation voltage and resistance | 3071 | $I_C$ = 10 mA dc; $I_B$ = 0.5 mA dc See 1.4. | $V_{CE(sat)}$ | | 0.15 | V dc |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 2 - Continued | | | | | | |
| Emitter to collector offset voltage | | $I_B$ = 200 µA dc; $I_E$ = 0 mA dc (see 4.5.5 and figure 9) | $V_{EC(ofs)1}$ | | | |
| 2N2432, 2N2432UB | | | | | 0.5 | mV dc |
| 2N2432A, 2N2432AUB | | | | | 0.4 | mV dc |
| Emitter to collector offset voltage | | $I_B$ = 1 mA dc; $I_E$ = 0 mA dc (see 4.5.5 and figure 9) | $V_{EC(ofs)2}$ | | | |
| 2N2432, 2N2432UB | | | | | 1.0 | mV dc |
| 2N2432A, 2N2432AUB | | | | | 0.7 | mV dc |
| Subgroup 3 | | | | | | |
| High-temperature operation: | | $T_A$ = +125°C | | | | |
| Collector to emitter cutoff current | 3041 | Bias condition C | $I_{CES2}$ | | | |
| 2N2432, 2N2432UB | | $V_{CE}$ = 25 V dc | | | 250 | nA dc |
| 2N2432A, 2N2432AUB | | $V_{CE}$ = 40 V dc | | | 250 | nA dc |
| Emitter to collector cutoff current | 3041 | Bias condition C; $V_{EC}$ = 15 V dc; $V_{BC}$ = 0 (see 4.5.2) | $I_{ECS2}$ | | 200 | nA dc |
| Low-temperature operation: | | $T_A$ = -55°C | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 10 µA dc; pulsed (see 4.5.1) | $h_{FE3}$ | 10 | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc; pulsed (see 4.5.1) | $h_{FE4}$ | 25 | | |
| Forward-current transfer ratio (inverted connection) | 3076 | $V_{CE}$ = 5 V dc; $I_E$ = 200 µA dc; pulsed (see 4.5.1 and 4.5.4) | $h_{FE(inv)2}$ | 1.8 | | |

See footnotes at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limit | Limit | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 4 | | | | | | |
| Small signal emitter collector on state resistance | | $I_B$ = 1 mA dc; $I_E$ = 0; $I_e$ = 100 μA ac (rms); f = 1 kHz (see 4.5.6 and figure 10) | $r_{ec(on)}$ | | | |
| 2N2432, 2N2432UB<br>2N2432A, 2N2432AUB | | | | | 20<br>15 | Ω<br>Ω |
| Small-signal short-circuit forward-current transfer ratio | 3306 | $V_{CE}$ = 5 V dc; $I_C$ = 1 mA dc; f = 20 MHz | $\|h_{fe}\|$ | 2 | 10 | |
| Open circuit output capacitance | 3236 | $V_{CB}$ = 0; $I_E$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{obo}$ | | 12.0 | pF |
| Input capacitance | 3240 | $V_{EB}$ = 0 V dc; $I_C$ = 0; 100 kHz ≤ f ≤ 1 MHz | $C_{ibo}$ | | 12.0 | pF |
| Subgroups 5, 6, and 7 | | | | | | |
| | | | | | | |
| Not applicable | | | | | | |
| | | | | | | |

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not applicable for JANS devices.
5/ Not required for laser marked devices.
* 6/ For end-point measurements, this test is required for the following subgroups:
Group B, subgroup 3, 4, and 5 (JANS).
Group B, step 1 (TX and TXV).
Group C, subgroup 2 and 6.



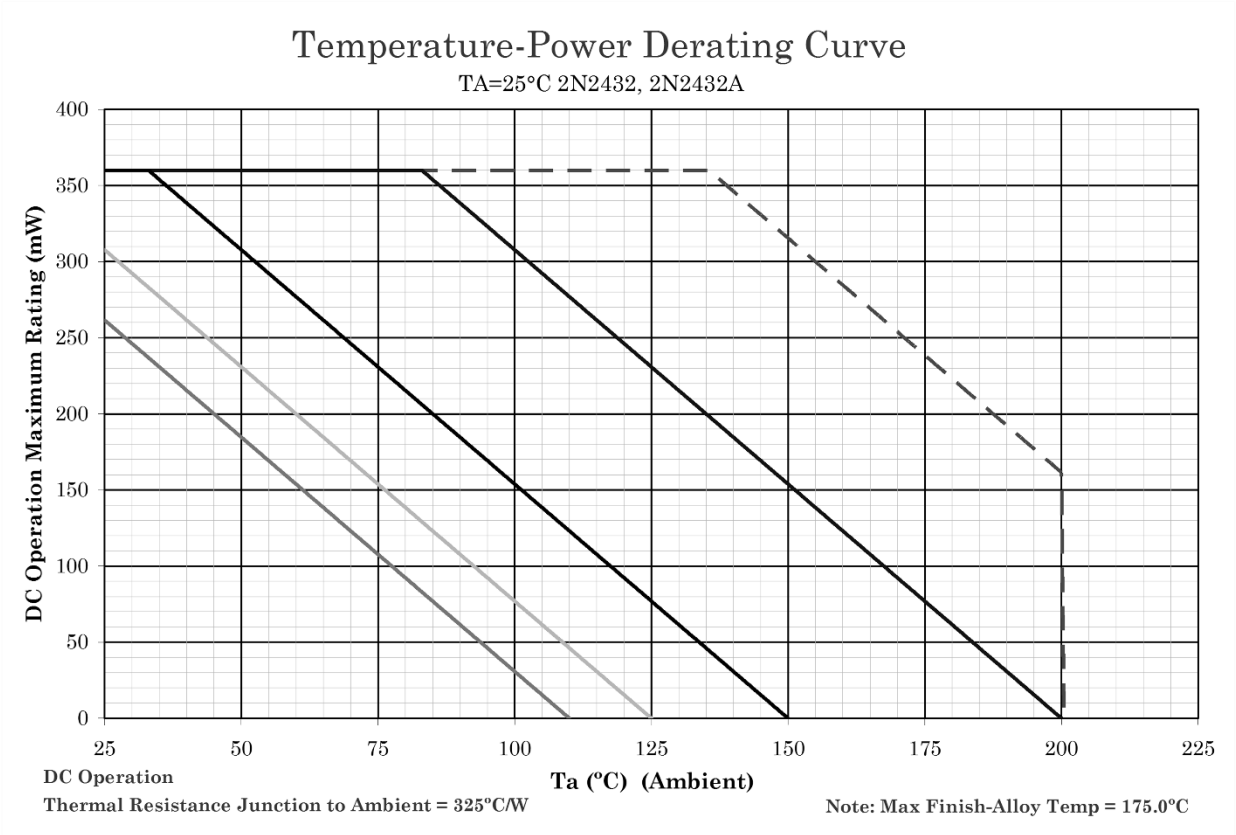
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TABLE II. Group E inspection (all quality levels) - for qualification only.

| Inspection | MIL-STD-750 | | Qualification |
|---|---|---|---|
| | Method | Conditions | |
| Subgroup 1 | | | 45 devices<br>c = 0 |
| Temperature cycling (air to air) | 1051 | Test condition C, 500 cycles | |
| Hermetic seal | 1071 | | |
| Fine leak<br>Gross leak | | | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.7 herein. | |
| Subgroup 2 | | | 45 devices<br>c = 0 |
| Intermittent life | 1037 | $V_{CB}$ = 10 V dc, 6,000 cycles, forced air cooling allowed on cooling cycle only. | |
| Electrical measurements | | See table I, subgroup 2 and 4.5.7 herein. | |
| Subgroup 4 | | | |
| Thermal resistance | 3131 | $R_{\theta JSP}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). | 15 devices,<br>c = 0 |
| Thermal impedance curves | | See table E-IX of MIL-PRF-19500, group E, subgroup 4. | Sample size<br>N/A |
| Subgroup 5 | | | |
| Not applicable | | | |
| Subgroup 6 | | | 11 devices |
| ESD | 1020 | | |
| Subgroup 8 | | | 45 devices<br>c = 0 |
| Reverse stability | 1033 | Condition B | |



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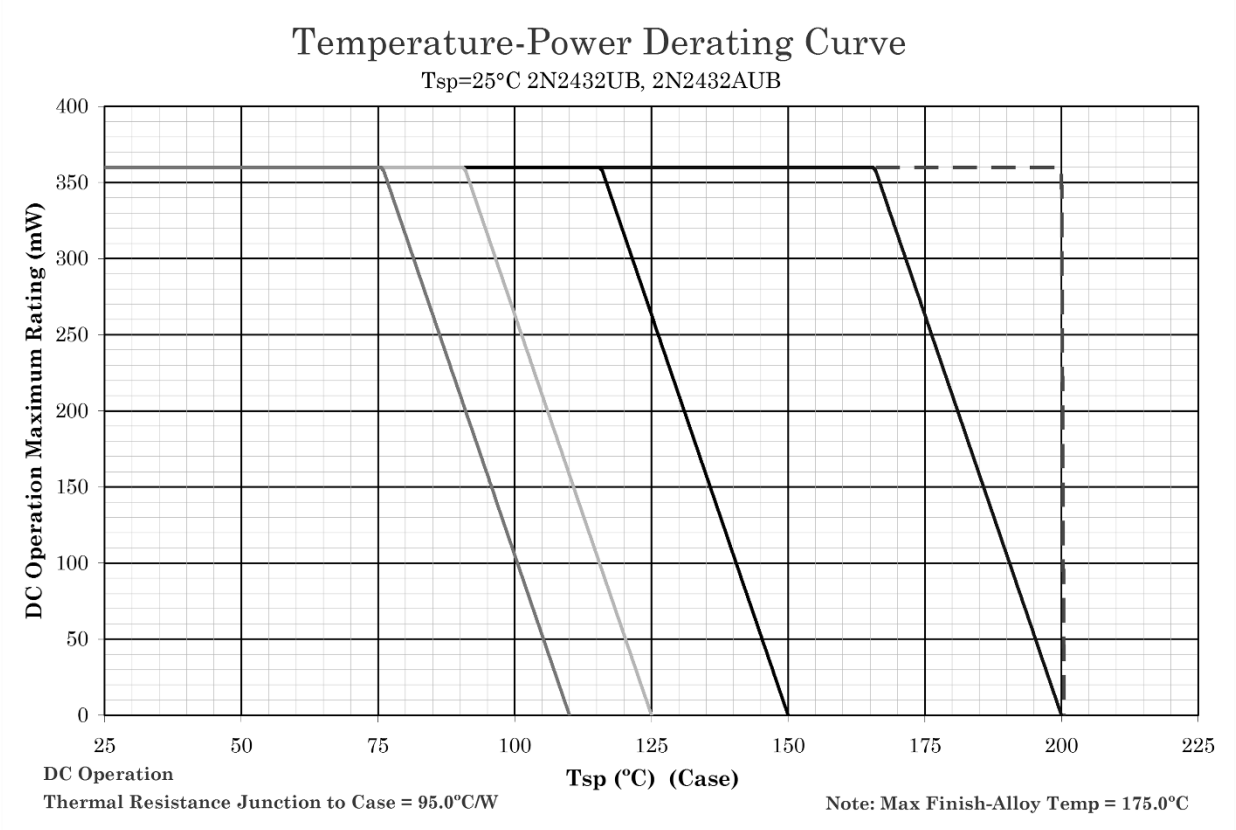

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 4. Derating for all devices ($R_{\theta JA}$) for 2N2432 and 2N2432A.



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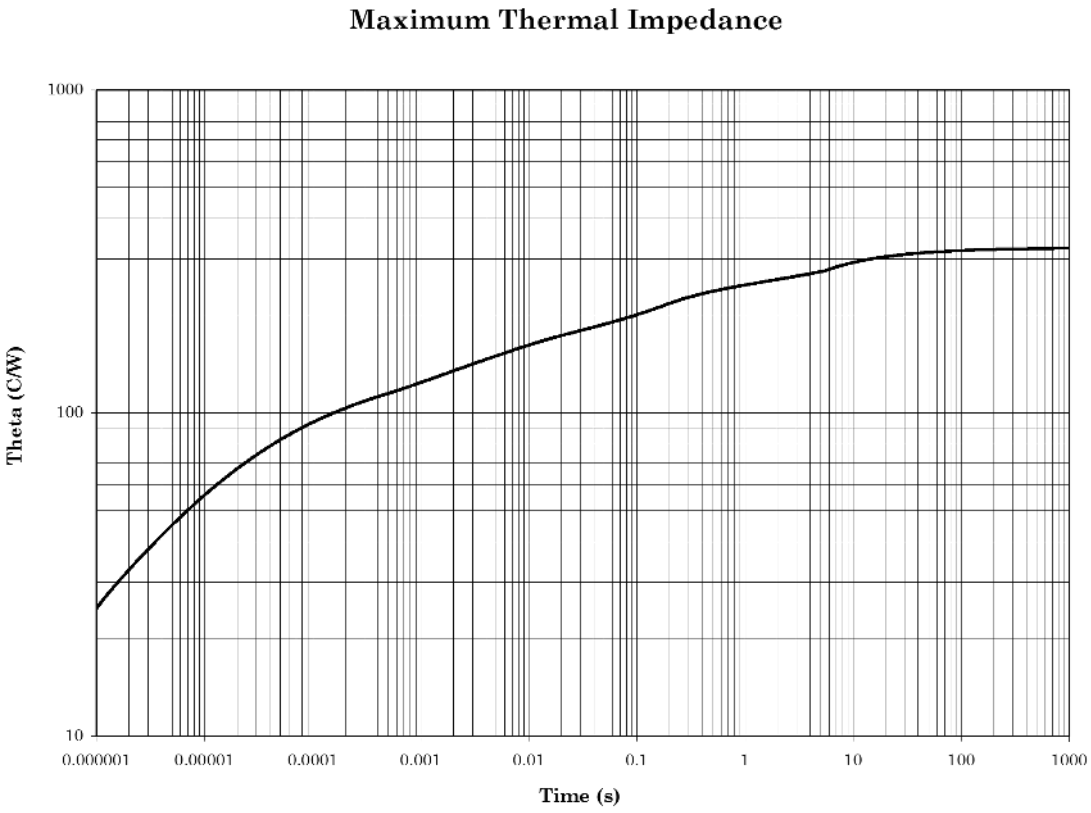

NOTES:

1. Top curve is thermal runaway loci and cannot be used as a derate design curve since it exceeds the maximum ratings for this part. Operating under this curve using these mounting conditions assures the device will not have a thermal runaway. This is the true inverse of the worst case thermal resistance value extrapolated out to the thermal runaway point.
2. Derate design curve constrained by the maximum junction temperatures and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq +150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq +125°C$, and +110°C to show power rating where most users want to limit $T_J$ in their application.

FIGURE 5. Derating for all devices ($R_{\theta JA}$) for 2N2432UB and 2N2432AUB.



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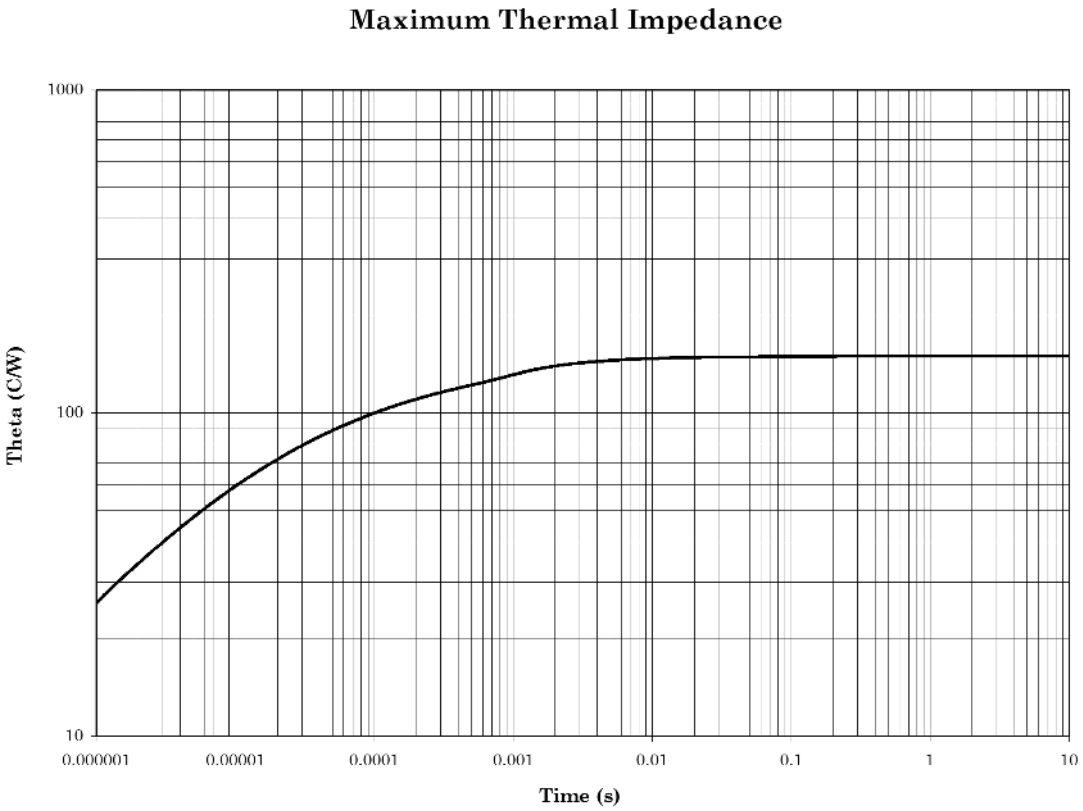



Resistance $R_{\theta JA}$ = 325°C/W.

FIGURE 6. Thermal impedance graph ($R_{\theta JA}$) for 2N2432 and 2N2432A.



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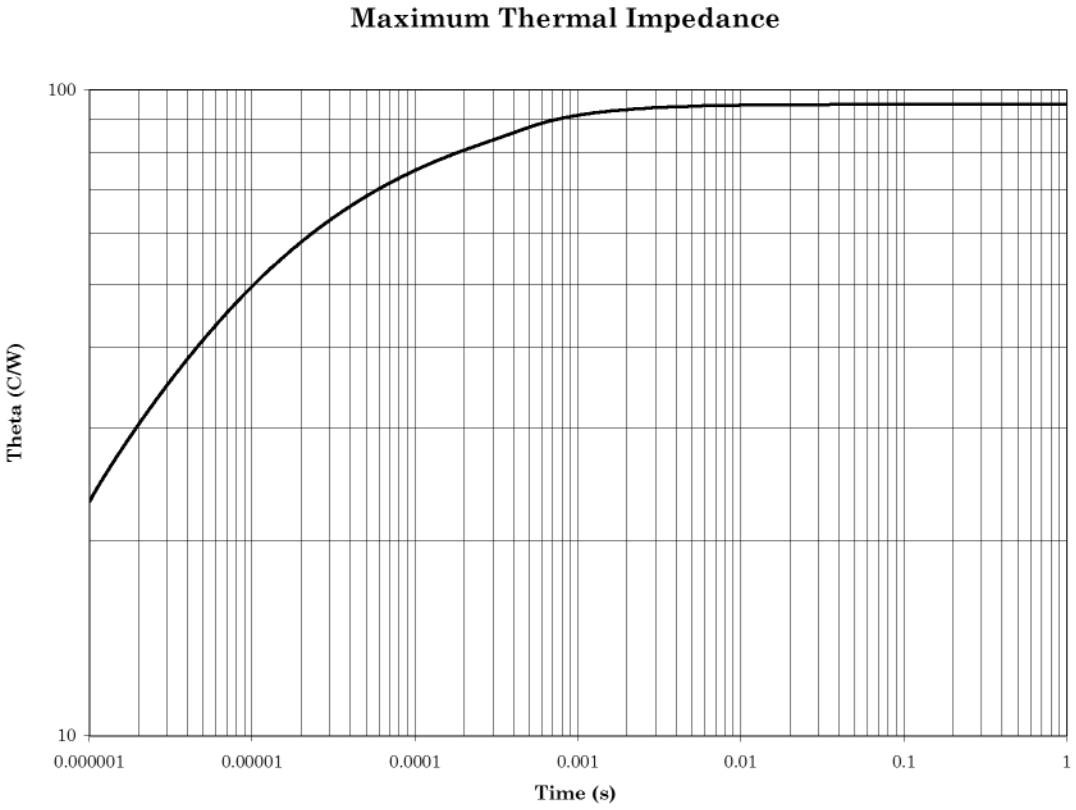



Resistance $R_{\theta JC}$ = 150°C/W.

FIGURE 7. Thermal impedance graph ($R_{\theta JC}$) for 2N2432 and 2N2432A.



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Resistance $R_{\theta JSP}$ = 95°C/W.

FIGURE 8. Thermal impedance graph ($R_{\theta JSP}$) for 2N2432UB and 2N2432AUB.



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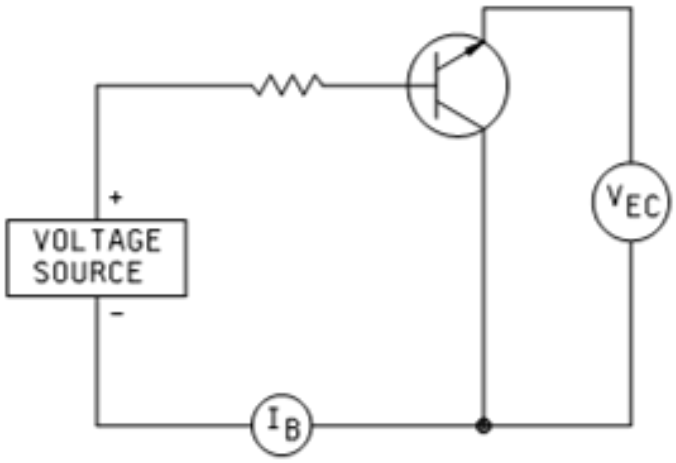



FIGURE 9. Emitter to collector offset voltage test circuit.

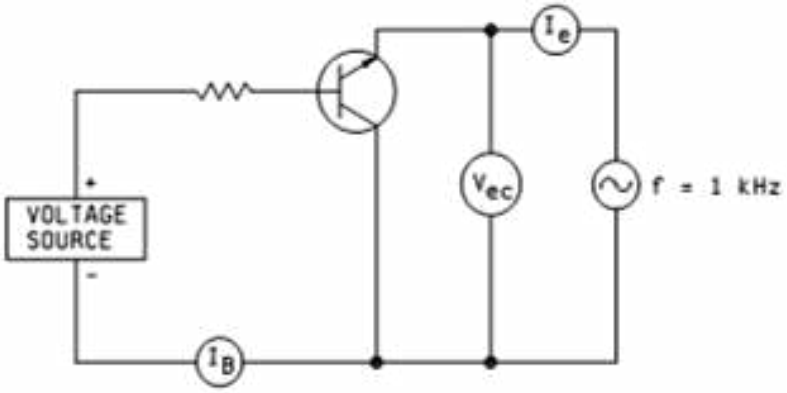



FIGURE 10. Small-signal emitter-collector on-state resistance test circuit.



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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete Part or Identifying Number (PIN), see title and section 1.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCB2N2434A) will be identified on the QML.

| JANC ordering information | |
|---|---|
| PIN | Manufacturer |
| | 43611 |
| 2N2432 | JANHCB2N2432A |
| 2N2432A | JANKCB2N2432A |

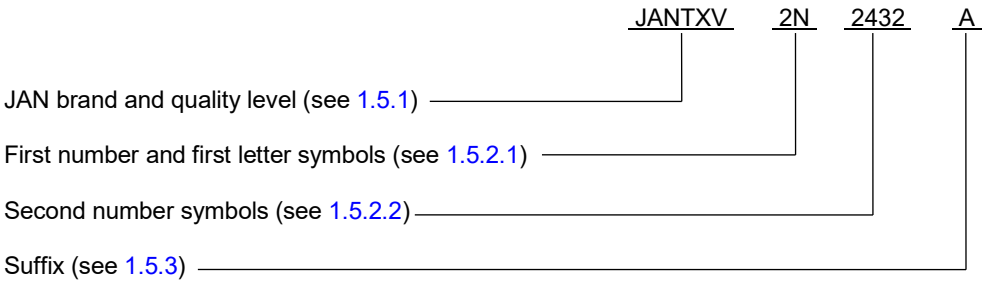
NOTE: The previous JANHC and JANKC (A version) was never qualified and has been removed in MIL-PRF19500/313J.



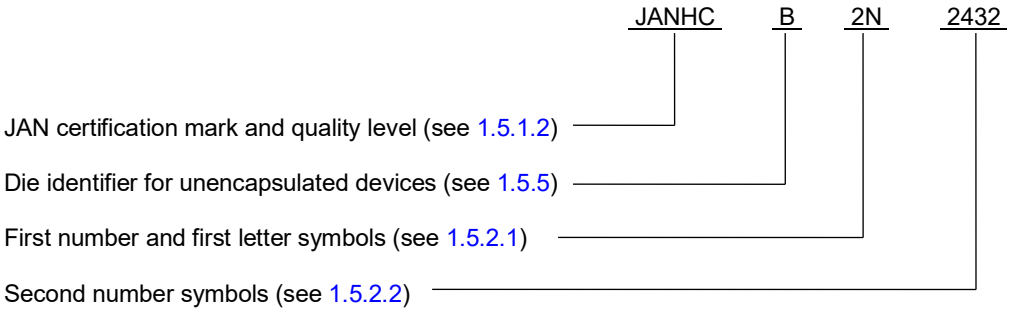
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6.5 PIN construction example. The PINs for encapsulated devices are construction using the following form.

6.5.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.




6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.




6.6 List of PINs. The following is a list of possible PINs available on this specification sheet.

PINs for types 2N2432, 2N2432A, 2N2432UB, and 2N2432AUB

| | | | |
|---|---|---|---|
| JAN2N2432 | JANTX2N2432 | JANTXV2N2432 | JANS2N2432 |
| JAN2N2432A | JANTX2N2432A | JANTXV2N2432A | JANS2N2432A |
| JAN2N2432UB | JANTX2N2432UB | JANTXV2N2432UB | JANS2N2432UB |
| JAN2N2432AUB | JANTX2N2432AUB | JANTXV2N2432AUB | JANS2N2432AUB |
| | JANHC2N2432 | JANKC2N2432A | |



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* 6.7 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army - CR
* Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

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